

4.1.4

List of packing instructions

4.1.4.1

Packing instructions concerning the use of packagings (except IBCs and large packagings)

P001		PACKING INSTRUCTION (LIQUIDS)			P001
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met.					
Combination packagings		Maximum capacity/net mass (see 4.1.3.3)			
Inner packagings		Outer packagings	Packing group I	Packing group II	Packing group III
Glass	10 L	Drums			
Plastics	30 L	steel (1A1, 1A2)	75 kg	400 kg	400 kg
Metal	40 L	aluminium (1B1, 1B2)	75 kg	400 kg	400 kg
		other metal (1N1, 1N2)	75 kg	400 kg	400 kg
		plastics (1H1, 1H2)	75 kg	400 kg	400 kg
		plywood (1D)	75 kg	400 kg	400 kg
		fibre (1G)	75 kg	400 kg	400 kg
		Boxes			
		steel (4A)	75 kg	400 kg	400 kg
		aluminium (4B)	75 kg	400 kg	400 kg
		other metal (4N)	75 kg	400 kg	400 kg
		natural wood (4C1, 4C2)	75 kg	400 kg	400 kg
		plywood (4D)	75 kg	400 kg	400 kg
		reconstituted wood (4F)	75 kg	400 kg	400 kg
		fibreboard (4G)	75 kg	400 kg	400 kg
		expanded plastics (4H1)	40 kg	60 kg	60 kg
		solid plastics (4H2)	75 kg	400 kg	400 kg
		Jerricans			
		steel (3A1, 3A2)	60 kg	120 kg	120 kg
		aluminium (3B1, 3B2)	60 kg	120 kg	120 kg
		plastics (3H1, 3H2)	30 kg	120 kg	120 kg
Single packagings					
Drums					
steel, non-removable head (1A1)			250 L	450 L	450 L
steel, removable head (1A2)			prohibited	250 L	250 L
aluminium, non-removable head (1B1)			250 L	450 L	450 L
aluminium, removable head (1B2)			prohibited	250 L	250 L
other metal, non-removable head (1N1)			250 L	450 L	450 L
other metal, removable head (1N2)			prohibited	250 L	250 L
plastics, non-removable head (1H1)			250 L*	450 L	450 L
plastics, removable head (1H2)			prohibited	250 L	250 L
Jerricans					
steel, non-removable head (3A1)			60 L	60 L	60 L
steel, removable head (3A2)			prohibited	60 L	60 L
aluminium, non-removable head (3B1)			60 L	60 L	60 L
aluminium, removable head (3B2)			prohibited	60 L	60 L
plastics, non-removable head (3H1)			60 L*	60 L	60 L
plastics, removable head (3H2)			prohibited	60 L	60 L
Composite packagings					
Plastics receptacle in steel, aluminium or plastics drum (6HA1, 6HB1, 6HH1)			250 L	250 L	250 L
Plastics receptacle in fibre or plywood drum (6HG1, 6HD1)			120 L*	250 L	250 L
Plastics receptacle in steel or aluminium crate or box or plastics receptacle in wood, plywood, fibreboard or solid plastics box (6HA2, 6HB2, 6HC, 6HD2, 6HG2 or 6HH2)			60 L*	60 L	60 L
Glass receptacle in steel, aluminium, fibre, plywood, expanded or solid plastics drum (6PA1, 6PB1, 6PG1, 6PD1, 6PH1 or 6PH2) or in a steel, aluminium, wood or fibreboard box or in a wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 or 6PD2)			60 L	60 L	60 L
Pressure receptacles, provided that the general provisions of 4.1.3.6 are met					

Not permitted for class 3, packing group I.

P001	PACKING INSTRUCTION (LIQUIDS) (continued)	P001
Special packing provisions:		
PP1	For UN Nos. 1133, 1210, 1263 and 1866 and for adhesives, printing inks, printing ink related materials, paints, paint related materials and resin solutions which are assigned to UN 3082, metal or plastics packagings for substances of packing groups II and III in quantities of 5 L or less per packaging are not required to meet the performance tests in chapter 6.1 when transported:	
	(a) in palletized loads, a pallet box or a unit load device, such as individual packagings placed or stacked and secured by strapping, shrink- or stretch-wrapping or other suitable means to a pallet. For sea transport, the palletized loads, pallet boxes or unit load devices shall be firmly packed and secured in closed cargo transport units. On roll-on/roll-off ships the unit loads may be carried in vehicles other than closed vehicles provided they are securely fenced to the full height of the cargo carried; or	
	(b) as an inner packaging of a combination packaging with a maximum net mass of 40 kg.	
PP2	For UN 3065, wooden barrels with a maximum capacity of 250 L and which do not meet the provisions of chapter 6.1 may be used.	
PP4	For UN 1774, packagings shall meet the packing group II performance level.	
PP5	For UN 1204, packagings shall be so constructed that explosion is not possible by reason of increased internal pressure. Gas cylinders and gas receptacles shall not be used for these substances.	
PP10	For UN 1791, for packing group II, the packaging shall be vented.	
PP31	For UN Nos. 1131, 1553, 1693, 1694, 1699, 1701, 2478, 2604, 2785, 3148, 3183, 3184, 3185, 3186, 3187, 3188, 3398 (PG II and III), 3399 (PG II and III), 3413 and 3414, packagings shall be hermetically sealed.	
PP33	For UN 1308, for packing groups I and II, only combination packagings with a maximum gross mass of 75 kg are allowed.	
PP81	For UN 1790 with more than 60% but not more than 85% hydrogen fluoride and UN 2031 with more than 55% nitric acid, the permitted use of plastics drums and jerricans as single packagings shall be two years from their date of manufacture	
PP93	For UN Nos. 3532 and 3534, packagings shall be designed and constructed to permit the release of gas or vapour to prevent a build-up of pressure that could rupture the packagings in the event of loss of stabilization.	

P002		PACKING INSTRUCTION (SOLIDS)			P002
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met.					
Combination packagings		Maximum net mass (see 4.1.3.3)			
Inner packagings		Outer packagings	Packing group I	Packing group II	Packing group III
Glass	10 kg	Drums			
Plastics ¹	30 kg	steel (1A1, 1A2)	125 kg	400 kg	400 kg
Metal	40 kg	aluminium (1B1, 1B2)	125 kg	400 kg	400 kg
Paper ^{1, 2, 3}	50 kg	other metal (1N1, 1N2)	125 kg	400 kg	400 kg
Fibre ^{1, 2, 3}	50 kg	plastics (1H1, 1H2)	125 kg	400 kg	400 kg
		plywood (1D)	125 kg	400 kg	400 kg
		fibre (1G)	125 kg	400 kg	400 kg
		Boxes			
		steel (4A)	125 kg	400 kg	400 kg
		aluminium (4B)	125 kg	400 kg	400 kg
		other metal (4N)	125 kg	400 kg	400 kg
		natural wood (4C1)	125 kg	400 kg	400 kg
		natural wood with sift-proof walls (4C2)	250 kg	400 kg	400 kg
		plywood (4D)	125 kg	400 kg	400 kg
		reconstituted wood (4F)	125 kg	400 kg	400 kg
		fibreboard (4G)	75 kg	400 kg	400 kg
		expanded plastics (4H1)	40 kg	60 kg	60 kg
		solid plastics (4H2)	125 kg	400 kg	400 kg
		Jerricans			
		steel (3A1, 3A2)	75 kg	120 kg	120 kg
		aluminium (3B1, 3B2)	75 kg	120 kg	120 kg
		plastics (3H1, 3H2)	75 kg	120 kg	120 kg
Single packagings					
Drums					
steel (1A1 or 1A2 ⁴)			400 kg	400 kg	400 kg
aluminium (1B1 or 1B2 ⁴)			400 kg	400 kg	400 kg
metal, other than steel or aluminium (1N1 or 1N2 ⁴)			400 kg	400 kg	400 kg
plastics (1H1 or 1H2 ⁴)			400 kg	400 kg	400 kg
fibre (1G ⁵)			400 kg	400 kg	400 kg
plywood (1D ⁵)			400 kg	400 kg	400 kg
⁴ These packagings shall not be used for substances of packing group I that may become liquid during transport (see 4.1.3.4).					
⁵ These packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).					

P002	PACKING INSTRUCTION (SOLIDS) <i>(continued)</i>			P002
The following packagings are authorized provided the general provisions of 4.1.1 and 4.1.3 are met.				
	Maximum net mass (see 4.1.3.3)			
Single packagings <i>(continued)</i>	Packing group I	Packing group II	Packing group III	
Jerricans				
steel (3A1 or 3A2 ⁴)	120 kg	120 kg	120 kg	
aluminium (3B1 or 3B2 ⁴)	120 kg	120 kg	120 kg	
plastics (3H1 or 3H2 ⁴)	120 kg	120 kg	120 kg	
Boxes				
steel (4A) ⁵	Not allowed	400 kg	400 kg	
aluminium (4B) ⁵	Not allowed	400 kg	400 kg	
other metal (4N) ⁵	Not allowed	400 kg	400 kg	
natural wood (4C1) ⁵	Not allowed	400 kg	400 kg	
natural wood with sift-proof walls (4C2) ⁵	Not allowed	400 kg	400 kg	
plywood (4D) ⁵	Not allowed	400 kg	400 kg	
reconstituted wood (4F) ⁵	Not allowed	400 kg	400 kg	
fibreboard (4G) ⁵	Not allowed	400 kg	400 kg	
solid plastics (4H2) ⁵	Not allowed	400 kg	400 kg	
Bags				
bags (5H3, 5H4, 5L3, 5M2) ⁵	Not allowed	50 kg	50 kg	
Composite packagings				
Plastics receptacle in steel, aluminium, plywood, fibre or plastics drum (6HA1, 6HB1, 6HG1 ⁵ , 6HD1 ⁵ , or 6HH1)	400 kg	400 kg	400 kg	
Plastics receptacle in steel or aluminium crate or box, wooden box, plywood box, fibreboard box or solid plastics box (6HA2, 6HB2, 6HC, 6HD2 ⁵ , 6HG2 ⁵ or 6HH2)	75 kg	75 kg	75 kg	
Glass receptacle in steel, aluminium, plywood or fibre drum (6PA1, 6PB1, 6PD1 ⁵ or 6PG1 ⁵) or in steel, aluminium, wood, or fibreboard box or in wickerwork hamper (6PA2, 6PB2, 6PC, 6PG2 ⁵ or 6PD2 ⁵) or in expanded or solid plastics packaging (6PH2 or 6PH1 ⁵)	75 kg	75 kg	75 kg	
⁴ These packagings shall not be used for substances of packing group I that may become liquid during transport (see 4.1.3.4).				
⁵ These packagings shall not be used when the substances being transported may become liquid during transport (see 4.1.3.4).				
Pressure receptacles, provided that the general provisions of 4.1.3.6 are met.				
Special packing provisions:				
PP7	For UN 2000, celluloid may be transported unpacked on pallets, wrapped in plastic film and secured by appropriate means, such as steel bands, as a single commodity in closed cargo transport units. Each pallet shall not exceed 1,000 kg.			
PP8	For UN 2002, packagings shall be so constructed that explosion is not possible by reason of increased internal pressure. Gas cylinders and gas receptacles shall not be used for these substances.			
PP9	For UN Nos. 3175, 3243 and 3244, packagings shall conform to a design type that has passed a leakproofness test at the packing group II performance level. For UN 3175 the leakproofness test is not required when the liquids are fully absorbed in solid material contained in sealed bags.			
PP11	For UN 1309, packing group III, and UN 1362, 5H1, 5L1 and 5M1 bags are allowed if they are overpacked in plastic bags and are wrapped in shrink or stretch wrap on pallets.			
PP12	For UN Nos. 2213 and 3077, 5H1, 5L1 and 5M1 bags are allowed when transported in closed cargo transport units.			
PP13	For articles classified under UN 2870, only combination packagings meeting the packing group I performance level are authorized.			
PP14	For UN Nos. 2211, 2698 and 3314, packagings are not required to meet the performance tests in chapter 6.1.			
PP15	For UN Nos. 1324 and 2623, packagings shall meet the packing group III performance level.			
PP20	For UN 2217, any sift-proof, tearproof receptacle may be used.			
PP30	For UN 2471, paper or fibre inner packagings are not permitted.			
PP31	For UN Nos. 1362, 1463, 1565, 1575, 1626, 1680, 1689, 1698, 1868, 1889, 1932, 2471, 2545, 2546, 2881, 3048, 3088, 3170, 3174, 3181, 3182, 3189, 3190, 3205, 3206, 3341, 3342, 3448, 3449 and 3450, packagings shall be hermetically sealed.			
PP34	For UN 2969 (as whole beans), 5H1, 5L1 and 5M1 bags are permitted.			
PP37	For UN Nos. 2590 and 2212, 5M1 bags are permitted. All bags of any type shall be transported in closed cargo transport units or be placed in closed rigid overpacks.			
PP38	For UN 1309, bags are permitted only in closed cargo transport units or as unit loads.			
PP84	For UN 1057, rigid outer packagings meeting the packing group II performance level shall be used. The packagings shall be designed and constructed and arranged to prevent movement, inadvertent ignition of the devices or inadvertent release of flammable gas or liquid.			
PP85	For UN Nos. 1748, 2208, 2880, 3485, 3486 and 3487, bags are not allowed.			
PP92	For UN Nos. 3531 and 3533, packagings shall be designed and constructed to permit the release of gas or vapour to prevent a build-up of pressure that could rupture the packagings in the event of loss of stabilization.			
PP100	For UN numbers 1309, 1323, 1333, 1376, 1435, 1449, 1457, 1472, 1476, 1483, 1509, 1516, 1567, 1869, 2210, 2858, 2878, 2968, 3089, 3096 and 3125, flexible, fibreboard or wooden packagings shall be sift-proof and water-resistant or shall be fitted with a sift-proof and water-resistant liner.			